CDA 4125: CONCEPTS OF PARALLEL AND DISTRIBUTED PROCESSING

# FINAL PROJECT SPECIFICATION

## 1. EXECUTIVE SUMMARY & RESEARCH OBJECTIVE

The objective of this final project is to conduct a rigorous empirical investigation comparing the computational efficiency, scaling behavior, and synchronization overhead of parallel programming paradigms on image processing kernels. Modern image workloads involve distinct memory and arithmetic patterns: spatial neighborhood convolutions (stencil computation) and spatial resampling (coordinate mapping and interpolation).

Each team will implement a reference sequential baseline, select **exactly one filtering operation** and **exactly one scaling operation**, and parallelize both operations using **exactly two distinct parallel mechanisms**. The resulting implementations will be tested across four distinct image resolutions to assess compute density, cache performance, communication overhead, and runtime speedup.

## 2. PERMITTED OPERATIONS & ARCHITECTURAL SCOPE

To ensure controlled experimental conditions, operations are strictly restricted to single-channel 2D spatial convolution and interpolation. You may use C or C++ libraries that support this operation but you need to implement the parallel mechanism on your own.

| CATEGORY | PERMITTED OPERATIONS (CHOOSE 1) | ALGORITHMIC & BOUNDARY RULES |
|---|---|---|
| **Filtering** *(Choose 1)* | • 3×3 Box Blur (equal weight 1/9)<br>• 3×3 Gaussian Blur (kernel weights: [1,2,1; 2,4,2; 1,2,1] / 16)<br>• 3×3 Laplacian Sharpen (center 5, cross −1) | Apply zero-padding on external image borders ($I(x,y) = 0$ for out-of-bound indices). Preserve original dimensions. Renormalization of border weights is strictly forbidden. |
| **Scaling** *(Choose 1)* | • Bilinear Interpolation (4-point weighted)<br>• Nearest-Neighbor Resampling | Implement arbitrary scale factors: both downsampling (0.5×) and upsampling (2.0×). Clamp out-of-bound coordinate mappings to edge pixels. |

## 3. PARALLEL MECHANISMS UNDER TEST (SELECT EXACTLY TWO)

Student teams must contrast the sequential baseline against **two** of the following four mechanisms:

- **OpenMP (Shared-Memory Directives):** Outer-loop row parallelization (`#pragma omp parallel for`) using static versus dynamic scheduling across worker thread teams ($p \in \{1, 2, 4, 8\}$).
- **POSIX Threads / Pthreads (Explicit Multi-Threading):** Manual thread dispatch, uniform row-range decomposition, explicit join synchronization, and thread barrier management.
- **CUDA (Massively Parallel GPU Architecture):** 2D execution grid with 16×16 or 32×32 thread blocks, explicit boundary masking, global DRAM access, and host-device transfer profiling.
- **MPI (Distributed Memory / Message Passing):** Domain decomposition of image sub-bands across discrete ranks, halo exchange or root-node scatter/gather communication, and IPC timing.

## 4. EXPERIMENTAL BENCHMARKING PROTOCOL & DATASETS

To evaluate scalability across workload sizes, all code variants must process an identical benchmark suite of 2 distinct photographic classes (portrait & landscape) at 2 fixed resolutions:

- **Resolution 1 (Small):** 512 × 512 pixels (0.26 Megapixels) — measures thread launch overhead.
- **Resolution 2 (Medium):** 1024 × 1024 pixels (1.05 Megapixels) — tests cache capacity limits.

**Timing Boundaries:** Execute 1 untimed warm-up run followed by 5 recorded repetitions per configuration. Report the median elapsed time and range.

Exclude disk I/O and display conversions from timing boundaries. For CUDA, report both kernel-only execution and host-to-device transfer-inclusive execution times.

## 5. FINAL TECHNICAL REPORT REQUIREMENTS

| CRITERION | EVALUATION STANDARD | POINTS |
|---|---|---|
| **Correctness & Accuracy** | Image should not be clipped and applicable both of the related process independently | 20 pts |
| **Implementation Quality** | Idiomatic code in both chosen mechanisms; absence of race conditions, deadlocks, and false sharing. | 25 pts |
| **Benchmarking Rigor** | Strict timing isolation; 5 trials per case; median/spread reporting; proper GPU transfer isolation. | 20 pts |
| **Written Report & Analysis** | High-density 4–6 page analysis. | 20 pts |
| **Reproducibility & Defense** | Be prepare to showcase your results in class including different images beyond your test scenaros | 15 pts |